

EQUIVARIANCE VERSUS FIRM NON-EXPANSIVENESS IN INCREMENTAL PROXIMAL GRADIENT METHODS FOR PLUG-AND-PLAY DENOISING

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ABSTRACT

Equivariant Plug-and-Play (PnP) methods symmetrize a learned denoiser by averaging its action over a group of image transformations, and rely on a stochastic Monte Carlo approximation for efficiency. We show that when the denoiser is firmly non-expansive (FNE), this stochastic scheme is not merely an unbiased estimator: it is closely related to Bertsekas incremental proximal-subgradient methods, since each randomly conjugated component is itself the resolvent of a maximal monotone operator. This gives direct access to Bertsekas convergence guarantees in the special case where the operator derives from a convex function. We carry out experiments on an SA-UNet denoiser trained with and without an FNE penalty and with and without equivariance data augmentation. We observe that FNE is the primary driver of both improved equivariance error and reconstruction quality, with augmentation playing a secondary role. Performance nevertheless improves further when the two are combined.

Index Terms— Plug-and-Play, equivariance, firmly non-expansive denoiser, proximal methods, image restoration, monotone operators

1. INTRODUCTION

Linear inverse problems in imaging aim to recover an image $\bar{\mathbf{x}} \in \mathbb{R}^N$ from degraded measurements $\mathbf{z} = \mathbf{A}\bar{\mathbf{x}} + \mathbf{e}$, where $\mathbf{A}$ models the acquisition operator (blur, subsampling, ...) and $\mathbf{e}$ is the measurement noise. Since these problems are generally ill-posed, solving them requires some form of regularization, classically an explicit penalty term added to the data-fidelity term. *Plug-and-Play* (PnP) methods offer an alternative: they replace this explicit regularizer with a pretrained denoiser $\mathbf{D}$, plugged into an iterative scheme that alternates between a data consistency step and a proximal step imposing regularity to the solution.

This strategy, originally heuristic [1, 2, 3], has since been given rigorous theoretical grounding whenever the denoiser satisfies certain stability properties, in particular firm non-expansiveness (FNE) [4, 5], which guarantees that $\mathbf{D}$ coincides with the resolvent of a maximal monotone operator [6] and places PnP squarely within the classical theory of proximal splitting methods [7, 8]. Although imposing Lipschitz

(or FNE) properties to a neural network is an NP-hard problem [9, 10] some numerical approaches have been shown to provide reasonable approximations [11, 12].

A well-known shortcoming of learned denoisers is that they are not equivariant to the natural symmetries of images (rotations, reflections): the same content, shown in a different orientation, is not necessarily denoised in a consistent way, which can introduce a directional bias in the reconstruction [13, 14]. Terris et al. [15] address this by symmetrizing the action of the denoiser, averaging it over a finite group of transformations. They also introduce a cheaper stochastic variant of this symmetrization, which they justify as an unbiased Monte Carlo estimator of the averaged operator. A convergence analysis of the resulting algorithm was provided in [16], under the assumption that the denoiser $\mathbf{D}$ can be expressed as $\mathbf{D} = \text{prox}_g$ for some weakly convex function g . This analysis controls the residuals of the iterates and shows that they converge close to critical points. No such analysis has yet been provided within the FNE formalism, which is precisely the focus of this contribution.

Equivariance and the FNE constraint are usually studied separately in the literature, and it remains debated which of the two impacts more reconstruction quality.

Contributions. In this paper, we give a second view of the stochastic equivariance scheme introduced by Terris et al.: rather than treating it as a simple stochastic estimator of an averaged operator, we recast it as an incremental algorithm which coincides with Bertsekas incremental proximal-subgradient method [17] under suitable assumptions on the denoiser. We show that, when the denoiser is FNE, the equivalence is exact: each randomly drawn component at every iteration is itself the resolvent of a maximal monotone operator (Lemma 1). Moreover, in the special case where this operator derives from a convex function, it directly inherits from Bertsekas convergence guarantees (Proposition 1). We then apply these results in experimental settings, where we train several variants of the same SA-UNet denoiser [18] (standard, FNE constrained, trained with equivariance, or combining both) to identify scenarios where equivariance brings a meaningful gain, and better understand which mechanism, equivariance or the FNE constraint, actually dominates.

2. EQUIVARIANT DENOISING AND AVERAGING

Plug-and-Play methods build on the idea to replace the proximal operator of an explicit regularizer with a generic denoiser $\mathbf{D}$. Concretely, they rely on the following iterative scheme:

$$(\forall k \in \mathbb{N}) \quad \mathbf{x}^{[k+1]} = \mathbf{D}(\mathbf{x}^{[k]} - \tau \mathbf{A}^\top (\mathbf{A} \mathbf{x}^{[k]} - \mathbf{z})). \quad (1)$$

Equivariance and averaging over the group – Terris et al. [15] propose to symmetrize the denoiser action by averaging it over a finite group $\mathcal{G}$ of transformations acting orthogonally on $\mathbb{R}^N$ through a representation $\mathbf{T}_\ell$ with $\ell \in \mathcal{G}$. In this work, we will focus on the group $\mathcal{G}$ that denotes the dihedral group of order 8 describing the symmetries of a square (the 4 rotations $\{0^\circ, 90^\circ, 180^\circ, 270^\circ\}$ combined with reflections), which naturally matches the symmetries of a discrete image on a square grid. This approach has since been revisited and extended in later work on equivariant denoising [16]. The equivariant denoising operator is defined as:

$$(\forall \mathbf{x} \in \mathbb{R}^N) \quad \mathcal{D}_\mathcal{G}(\mathbf{x}) = \frac{1}{|\mathcal{G}|} \sum_{\ell \in \mathcal{G}} \mathbf{T}_\ell^{-1} \mathbf{D}(\mathbf{T}_\ell \mathbf{x})$$

and the equivariant PnP iteration replaces the denoising step of Algorithm (1) with $\mathcal{D}_\mathcal{G}$:

$$(\forall k = 0, 1, \dots) \quad \mathbf{x}^{[k+1]} = \mathcal{D}_\mathcal{G}(\mathbf{x}^{[k]} - \tau \mathbf{A}^\top (\mathbf{A} \mathbf{x}^{[k]} - \mathbf{z})).$$

Monte Carlo approximation – Computing $\mathcal{D}_\mathcal{G}$ requires $|\mathcal{G}|$ calls to the denoiser at every iteration, which may become expensive. Terris et al. [15] therefore propose a stochastic version, where a single group element is drawn at each iteration instead of averaging over all of them:

For $k = 0, 1, \dots$

$$\left| \begin{array}{l} \text{Select } \ell_k \sim \mathcal{U}(\mathcal{G}) \text{ independently of the previous draws} \\ \mathbf{x}^{[k+1]} = \mathbf{T}_{\ell_k}^{-1} \mathbf{D}(\mathbf{T}_{\ell_k}(\mathbf{x}^{[k]} - \tau \mathbf{A}^\top (\mathbf{A} \mathbf{x}^{[k]} - \mathbf{z}))) \end{array} \right. \quad (2)$$

The authors show experimentally that enforcing equivariance of the denoiser to certain groups of transformations, using this Equivariant-PnP algorithm, strongly improves both the stability of the algorithm and the quality of reconstruction. However, a deeper theoretical investigation is still needed and is provided in the next section.

3. A BERTSEKAS FORMULATION OF THE MONTE CARLO EQUIVARIANT PNP

We now give a new interpretation of Algorithm (2), which no longer treats it as a simple stochastic estimator of $\mathcal{D}_\mathcal{G}$, but instead as an instance of Bertsekas incremental proximal-subgradient methods [17]. This re-interpretation allows us to benefit from the well-established convergence results in some specific settings.

FNE and orthonormal transforms – Suppose that $\mathbf{D}$ is firmly non-expansive, defined on the whole space. By Minty's

theorem [6], there then exists a maximal monotone operator $\mathbf{M}: \mathbb{R}^N \rightrightarrows \mathbb{R}^N$ such that $\mathbf{D} = \mathbf{J}_\mathbf{M} := (\text{Id} + \mathbf{M})^{-1}$, where $\mathbf{J}_\mathbf{M}$ denotes the resolvent of $\mathbf{M}$. The following lemma, a classical result in monotone operator theory (see [7, 8]), shows that this structure is preserved under orthogonal conjugation:

Lemma 1 (Invariance of the resolvent under orthogonal conjugation). *Let $\mathbf{M}$ be a maximal monotone operator on $\mathbb{R}^N$, and let $\mathbf{T}_\ell$ be an orthogonal transformation. Then*

$$\mathbf{J}_{\widetilde{\mathbf{M}}_\ell} = \mathbf{T}_\ell^{-1} \circ \mathbf{J}_\mathbf{M} \circ \mathbf{T}_\ell \quad \text{where} \quad \widetilde{\mathbf{M}}_\ell := \mathbf{T}_\ell^{-1} \mathbf{M} \mathbf{T}_\ell,$$

and $\widetilde{\mathbf{M}}_\ell$ is also a maximal monotone operator.

So for every $\ell \in \mathcal{G}$, the term $\mathbf{T}_\ell^{-1} \mathbf{D} \mathbf{T}_\ell$ used in Algorithm (2) is itself exactly the resolvent of a maximal monotone operator $\widetilde{\mathbf{M}}_\ell := \mathbf{T}_\ell^{-1} \mathbf{M} \mathbf{T}_\ell$. We can therefore rewrite the algorithm in a form that implies a *randomized resolvent interpretation* involving maximal monotone operators :

For $k = 0, 1, \dots$

$$\left| \begin{array}{l} \text{Select } \ell_k \sim \mathcal{U}(\mathcal{G}) \text{ independently of the previous draws} \\ \widetilde{\mathbf{M}}_{\ell_k} := \mathbf{T}_{\ell_k}^{-1} \mathbf{M} \mathbf{T}_{\ell_k} \\ \mathbf{x}^{[k+1]} = \mathbf{J}_{\widetilde{\mathbf{M}}_{\ell_k}}(\mathbf{x}^{[k]} - \tau \mathbf{A}^\top (\mathbf{A} \mathbf{x}^{[k]} - \mathbf{z})). \end{array} \right. \quad (3)$$

This change of viewpoint that consists in replacing a *randomly orthogonal conjugated resolvent* with the *resolvent of a random operator* $\widetilde{\mathbf{M}}_{\ell_k}$ drawn from a finite family, is exactly what lets us step directly into Bertsekas framework.

Special case: $\mathbf{M}$ subdifferential of a convex function – When the maximal monotone operator is a convex subdifferential, we show that the method becomes a randomized incremental proximal method of Bertsekas [17]. The case where the denoiser is associated with an explicit regularizer corresponds to $\mathbf{M} = \tau \partial \phi$, for some proper closed convex function $\phi: \mathbb{R}^N \rightarrow \mathbb{R}$ and $\tau > 0$. The resolvent $\mathbf{J}_\mathbf{M}$ is then exactly the proximal operator:

$$(\forall \mathbf{x} \in \mathbb{R}^N) \quad \mathbf{D}(\mathbf{x}) = \mathbf{J}_{\tau \partial \phi}(\mathbf{x}) = \text{prox}_{\tau \phi}(\mathbf{x})$$

In that specific case, $\mathbf{D}$ is FNE and it fits exactly the setting of [17] where each $\ell \in \mathcal{G}$ corresponds to a component $\phi_\ell := \phi \circ \mathbf{T}_\ell$ (also convex, lower semicontinuous, and proper, since $\mathbf{T}_\ell$ is an isometry), and the incremental iteration alternates a gradient step on the data-term with a proximal step on the randomly drawn component ϕ_{ℓ_k} :

For $k = 0, 1, \dots$

$$\left| \begin{array}{l} \text{Select } \ell_k \sim \mathcal{U}(\mathcal{G}) \text{ independently of the previous draws} \\ \phi_{\ell_k} = \phi \circ \mathbf{T}_{\ell_k} \\ \mathbf{x}^{[k+1]} = \text{prox}_{\tau_k \phi_{\ell_k}}(\mathbf{x}^{[k]} - \tau_k \mathbf{A}^\top (\mathbf{A} \mathbf{x}^{[k]} - \mathbf{z})). \end{array} \right. \quad (4)$$

Building on this framework, we can consider Bertsekas convergence results in terms of function value to derive the

following statement, specified to our setting, in the fixed-step-size case.

Proposition 1 (Fixed-step objective-value bound). *For every $\mathbf{x} \in \mathbb{R}^N$, let $F(\mathbf{x}) := \frac{1}{2}\|\mathbf{Ax} - \mathbf{z}\|_2^2 + L^{-1} \sum_{\ell=1}^L \phi(\mathbf{T}_\ell \mathbf{x})$ with $\mathbf{z} \in \mathbb{R}^M$, $\mathbf{A} \in \mathbb{R}^{M \times N}$, $\phi: \mathbb{R}^N \rightarrow \mathbb{R}$ a convex, lower-semicontinuous, and proper function, and $(\mathbf{T}_\ell)_{1 \leq \ell \leq L}$ a family of orthonormal transforms. Let $(\mathbf{x}^{[k]})_{k \in \mathbb{N}}$ be the sequence generated by Algorithm (4) with a constant step size $\tau_k \equiv \tau$, and $F^* := \inf_{\mathbf{x} \in \mathbb{R}^N} F(\mathbf{x})$ and assume that there exists $c \in \mathbb{R}$ such that, with probability 1, $\max \left\{ \left\| \nabla \left(\frac{1}{2} \|\mathbf{Ax}^{[k]} - \mathbf{z}\|_2^2 \right) \right\|, \|\partial \phi(\mathbf{T}_{\ell_k} \mathbf{x}^{[k]})\| \right\} \leq c$, for every $\ell_k \in \mathcal{G}$. Then, with probability 1:*

- (i) if $F^* = -\infty$, then $\inf_{k \geq 0} F(\mathbf{x}^{[k]}) = -\infty$;
- (ii) if $F^* > -\infty$, then $\inf_{k \geq 0} F(\mathbf{x}^{[k]}) \leq F^* + \frac{5\tau c^2}{2}$.

In other words, with a fixed step size, the sequence does not converge exactly to F^* , but the result guarantees that the iterates attain objective values within an $O(\tau)$ neighborhood of the optimum. The error bound decreases linearly with τ .

4. EXPERIMENTS

Our experiments are run with Deep Inverse project library [19] on a standard restoration problem where $\mathbf{A}$ denotes a Levin blur kernel and $\mathbf{e}$ a white Gaussian noise with standard deviation set to $\sigma = 0.03$. We run Algorithm (2) with several denoisers that share the same architecture but differ in training, depending on whether the FNE property and/or equivariance are enforced during training. Our main goal is to evaluate the impact of using an FNE denoiser within Algorithm (2). Note that Proposition 1 guarantees error bound decreasing only when in the convex regularization setting, and its extension to the FNE case has not been established; we therefore restrict our analysis to an empirical evaluation.

Choice of denoiser: SA-UNet – Throughout our experiments, we use a SA-UNet (*Spatial Attention U-Net*) [18], an architecture originally proposed for retinal vessel segmentation, which we adapt here to image denoising. SA-UNet follows the classic encoder–decoder structure of U-Net, with skip connections between matching levels of the encoder and decoder.

Training – The denoisers are trained on a subset of ImageNet containing 50,000 images, split into 45×45 patches. Additive Gaussian noise at level $\sigma = 0.03$ is used to generate noisy observations $\mathbf{y} = \bar{\mathbf{x}} + \mathbf{n}$ where $\mathbf{n} \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$. The training follows the protocol proposed by Pesquet et al. [4], which adds to the denoising loss a penalty that encourages the denoiser to be firmly non-expansive. Writing $\mathbf{D} := \mathbf{D}_\theta$ for the denoiser parameterized by θ and $\mathbf{Q}_\theta := 2\mathbf{D}_\theta - \text{Id}$, the loss function is

$$\mathcal{L}(\theta) = \sum_{i=1}^B \frac{\|\mathbf{D}_\theta(\mathbf{y}_i) - \mathbf{x}_i\|_2^2}{2BP^2} + \lambda \max(\|\mathbf{J}_{\mathbf{Q}_\theta}(\tilde{\mathbf{x}})\|_2 - (1 - \epsilon), 0)^2$$

where B is the batch size, P the patch size, $\lambda > 0$ controls the strength of the FNE constraint, $\epsilon > 0$ is a margin parameter, and $\tilde{\mathbf{x}}$ is the point at which the spectral norm of the

Jacobian $\mathbf{J}_{\mathbf{Q}_\theta}$ is evaluated. This penalty encourages $\mathbf{Q}_\theta$ to be non-expansive and, in turn, $\mathbf{D}_\theta$ to be firmly non-expansive [4]. We therefore consider four denoiser variants:

- **SA-UNet base**: standard training, with $\lambda = 0$;
- **SA-UNet equiv.**: training with data augmentation using the transformations of the group D_4 with $\lambda = 0$;
- **SA-UNet FNE**: training with the FNE (i.e, $\lambda > 0$);
- **SA-UNet FNE + equiv.**: combining the FNE penalty (i.e, $\lambda > 0$) with D_4 augmentation.

For FNE-based training, λ has been selected to minimize the spectral norm of the Jacobian of $\mathbf{Q}_\theta$. Optimization is carried out with Adam. Models with the FNE constraint are obtained by fine-tuning the corresponding denoiser, while the equivariant models are trained by taking the group $\mathcal{G}$ that denotes the dihedral group of order 8 describing the symmetries of a square (the 4 rotations $\{0^\circ, 90^\circ, 180^\circ, 270^\circ\}$ combined with reflections).

Table 1 reports, for the four variants of denoiser and on a test patch of size 45×45 taken from a reference image, the average equivariance error $E_{\text{eq}}^{\text{moy}}$ (with its standard deviation E_{eq}^σ), together with the deviation from the non-expansiveness constraint σ_{FNE} (the spectral norm of the Jacobian of $\mathbf{Q}_\theta$, estimated with 50 iterations of the power method, which should ideally be no larger than 1) and its absolute and relative residuals.

Method	$E_{\text{eq}}^{\text{moy}}$	E_{eq}^σ	σ_{FNE}	Abs. res.	Rel. res.
SA-UNet base	0.015	0.006	10.080	1.377	0.022
SA-UNet Equiv.	0.005	0.002	2.198	0.455	0.007
SA-UNet FNE	0.003	0.001	0.994	0.310	0.005
SA-UNet FNE + Equiv.	0.002	0.001	0.994	0.344	0.005

Table 1. Denoiser performance.

Compared PnP variant – Using these denoisers, we evaluate six PnP implementations:

- **PnP**: Algorithm (1) with SA-UNet base;
Standard PnP procedure.
- **PnP-Equiv.**: Algorithm (1) with SA-UNet equiv;
Standard PnP procedure when the training involved equivariant augmentation.
- **PnP-FNE**: Algorithm (1) with SA-UNet FNE;
Implementation from [4].
- **PnP-FNE-Equiv**: Algo. (1) with SA-UNet FNE + Equiv;
Implementation from [4] when the training involved equivariant augmentation.
- **Bert-PnP**: Algorithm (2) with SA-UNet base;
Implementation provided in [15].
- **Bert-PnP-FNE** (Proposed): Algo.(2) with SA-UNet FNE.

Results: Importance of FNE – The convergence criteria (PSNR and iterate convergence profiles) are displayed for two images (Figure 1 and Figure 2). Both results are consistent with the theoretical properties discussed in Section 3. In particular, the FNE-regularized variants are the only tested methods for which the residual is observed to converge to zero. We can also notice that these three strategies PnP-FNE,

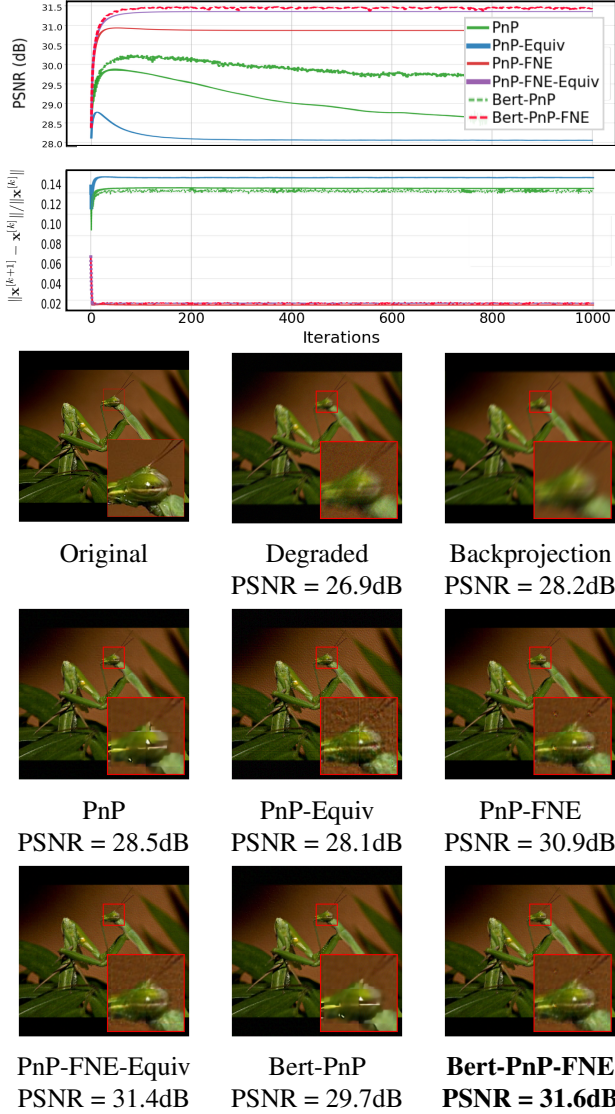


Fig. 1. (1st row) PSNR, (2nd row) Residual (Set of 9 images) Original/degraded and reconstruction results for Image 1.

PnP-FNE-Equiv, and Bert-PnP-FNE achieve the best PSNR performance.

Results: Stochastic Equivariant implementation – The PSNR curves, meanwhile, reveal notable differences between the strategies. Overall, equivariance applied at inference in the PnP procedure (i.e., Bert-PnP or Bert-PnP-FNE) brings clearer benefit than equivariance introduced during training (i.e., PnP-Equiv or PnP-FNE-Equiv). The benefit of equivariance at inference is discussed in more detail in [15].

Results: FNE and Equivariance – From the PSNR plots, both combining equivariance at inference (i.e., Bert-PnP-FNE) or at training time (i.e., PnP-FNE-Equiv) lead to better results than the standard FNE variant (i.e., PnP-FNE). Nonetheless, Bert-PnP-FNE remains slightly better than PnP-

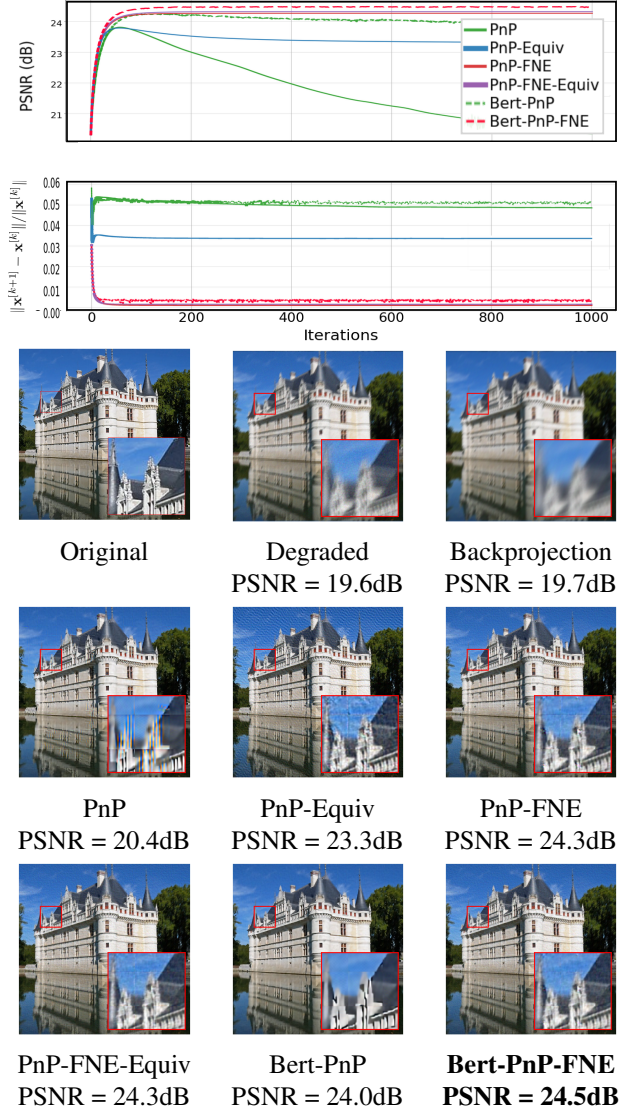


Fig. 2. (1st row) PSNR, (2nd row) Residual (Set of 9 images) Original/degraded and reconstruction results for Image 2.

FNE-Equiv, while also allowing the use of an FNE denoiser without requiring any retraining.

5. CONCLUSION

In this work, we showed that equivariant PnP with an FNE denoiser is closely related to Bertsekas incremental proximal-subgradient methods, yielding asymptotic guarantees in the convex case, and we confirmed experimentally in the considered setting, that FNE has the dominant effect of improved equivariance error and reconstruction quality, with data augmentation offering a complementary gain. As future work, extending our convergence proof beyond the convex case remains an open challenge, one that our experimental observations suggest is within reach.

6. REFERENCES

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